



Armada Kalamunda Group

Obesity in Pregnancy Guideline

1. Purpose

To identify and provide optimal care for pregnant women with an increased BMI to decrease risk and improve outcomes for pregnancy, birth and postnatally.

2. Applicability

Applies to maternity health care providers which includes but not limited to midwives, GP Obstetricians, specialist Obstetricians, anaesthetists and allied health professionals.

3. Guidelines

3.1 Background

- Obesity in pregnancy is now one of the most important challenges in obstetric care. Approximately 50 per cent of women who become pregnant are either overweight (BMI>25 – 30) or obese (BMI>30).
- There is substantial evidence that obesity in pregnancy contributes to increased morbidity and mortality for both mother and baby. The Confidential Enquiry into Maternal and Child Health (CEMACH) (2007) report indicated that more than half of the women who died from direct / indirect causes were obese.

Table 1: Health risks of obesity in pregnancy include but not limited to:

Antenatal	• Impaired fasting glucose and impaired glucose tolerance; gestational diabetes, three times more common in obese women • Miscarriage • Stillbirth, 2-3 fold increased risk for intrauterine fetal death • Hypertensive disorders in pregnancy, raised BMI increases risk of PET by 50% • Thromboembolism • Obstructive Sleep apnoea • Maternal death • Abnormalities in fetal growth and development, early neonatal death
Intra-partum	• Induction of labour, prolonged labour and failure to progress • Increased rate of instrumental delivery, caesarean section and postpartum haemorrhage • Shoulder dystocia • Difficulties with fetal heart rate monitoring • Difficulties with labour analgesia

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	• Use of general anaesthesia
Anaesthetic	• Difficulty with positioning • Difficulty with siting epidural or spinal anaesthetic and increased risk of dislodgement • Difficulty maintaining an adequate airway. • Increased risk of need for ICU care post operatively
Post-partum	• Delayed wound healing. • Increased rates of wound infection • Greater likelihood of needing support with breastfeeding establishment and continuation • Postnatal depression • Long term neonatal consequences: Neonatal body composition, Infant weight gain, Obesity, metabolic disorders in childhood

4. Antenatal management

All woman with booking BMI >40 at booking will be referred to FSH. Reassess BMI at 32-34 weeks. If BMI >38 at 32-34 weeks, a referral needs to be sent to FSH for appointment at 36 weeks

All antenatal women attending AHS should have their BMI (according to their pre-pregnancy weight or the earliest weight in pregnancy) done at the first visit.

- Routine scheduled antenatal visits may need to be adjusted to be more frequent according to the level of obesity and risk factors. Refer to Appendix 1- Antenatal Clinical Guideline for visit schedule. Provide woman with BLOOM Pack and/or RANZCOG brochure regarding “weight management during Pregnancy” at their first antenatal visit.
- Document woman’s weight and calculate BMI at each visit.
- Principles of weight management care: Sensitive language to reduce weight stigma:

BMI classification(kg/m ²)	GWG	Total GWG
Underweight <18.5	Trimester 1kg	Singleton kg
Normal 18.5-24.9*	All women 0.5-2.0	Normal 11.5-16
Overweight 25.0-34.9*	Trimester 2+3 kg/week	Overweight 7-11.5
Obese I 30.0-34.9*	Underweight 0.5	Obese 5-9
Obese II 35.0-39.9*	Normal 0.4	Twins kg
Obese III >40*	Overweight 0.3	Normal 17-25
	Obese 0.2	Overweight 14-23
*variation for Asian background		Obese 11-19

- Folate supplementation is known to prevent Neural Tube Defects. Recommend Folic acid 5 mg daily till the end of first trimester.
- All woman with BMI >35 with no other risk factors can be seen in GP Obstetric clinic.
- Women with BMI >35 with additional risk factors should be referred to Specialist Obstetrician early in pregnancy.
- All woman with an increased BMI should ideally be referred to a dietician (if dietician services present) or be appropriately counselled by GP Obstetrician or Specialist Obstetrician
- All women with BMI >40 or BMI >35 with comorbidities and planning to deliver at AHS to be referred to anaesthetist in PAC for counselling and discussion regarding regional/ general anaesthesia and have a plan ready for labour management. Anaesthetic referral should ideally be completed by 28-32 weeks.
- Women who have had Bariatric surgery require closer monitoring for nutritional deficiencies and monitoring of baby's growth. Studies have indicated a trend towards increased risk of SGA, IUGR and decreased birth weight. These women should be referred to the Specialist Clinic for review.
- Woman with BMI >35 and previous history of caesarean should be referred to the Specialist Obstetrician clinic to discuss risks and outcomes for TOLAC.
- Early screening for diabetes and other endocrinological abnormalities should be encouraged, preferably at the time of first antenatal attendance. If early diabetic screening is normal, repeat the screening at 24-28 weeks gestation.
- All woman with BMI >35 and woman with BMI>30 with comorbidities should have serial growth scans every 3 weeks. The estimated fetal weights must be plotted on customised GROW chart. Include BMI on request for ultrasound. These are besides the routine dating, FTS and anatomy scans.
- Have VTE assessment completed at first antenatal visit and thereafter if any risk factors develop. Consider antenatal thromboprophylaxis with LMWH for woman with two or more additional risk factors for [VTE](#).
- Encourage 30 minutes of low intensity exercise daily.
- Home birth and water birth is not recommended for woman with a BMI >35 refer to [WA Health Policy for Publicly Funded Homebirths including Guidance for Consumers, Health Professionals and Health Services](#).
- Women with BMI >40 refusing to go to FSH should be counselled about the risks and all discussions documented clearly in the patient case notes. A nonstandard management sticker needs to be placed into the patient notes with signature from the counselling specialist and the patient. Reason for noncompliance should also be documented.
- Patient handling shall comply with the [EMHS Bariatric Management Policy](#).

5. Intrapartum care

5.1 Vaginal birth

- For women wishing to have a TOLAC, the risks and statistics of having a successful VBAC must be discussed in a manner that supports shared decision making. Consider early anaesthetic consultation in labour.
- Obesity alone, in the absence of any other obstetric or medical indication, is not an indication for IOL. Individualise decision making about mode and timing of birth based on an assessment of potential risk factors for poor birth outcomes.
- Inform Labour Ward coordinator and GP Obstetrician on call when woman with BMI>35 arrives in labour.

- Ensure venous access with a large bore cannula when woman with high BMI arrive in labour.
- Collect blood for “Group and hold” when IV access is performed. Women with an increased BMI are at higher risks for PPH and intrapartum complications.
- Use appropriately sized BP cuff.
- Use Scan to confirm presentation if uncertain or difficult to palpate.
- Have low threshold for use of fetal scalp clip for fetal heart monitoring.
- Woman with BMI ≥ 35 should have continuous intrapartum fetal monitoring. Woman with BMI >30 with comorbidities should also have continuous fetal monitoring.
- Use TED stockings.
- Consider keeping oral intake to minimum and to clear fluids.
- Keep Anaesthetist on call informed.
- Early epidural placement should be considered.
- Maintain an awareness of the increased risk of shoulder dystocia and postpartum haemorrhage. Active third stage management is recommended for all obese women in labour.
- Consider the possible requirement of additional blood products, keep anaesthetist informed of the woman’s progress.
- For woman with BMI >40 or more assess pressure areas and maintain skin integrity.
- Ensure manual handling equipment is available and used.

5.2 Caesarean delivery

- Notify OT staff of woman’s BMI.
- Caesarean Section (CS) is technically more difficult.
- Risk of wound infection and excess blood loss is increased in high BMI women.
- Consider use of negative pressure dressings on closure to reduce fluid collection in the wound in woman with BMI ≥ 30 .
- All women with high BMI require Pharmacological VTE prophylaxis post elective or emergency CS unless contraindicated. This should be continued for seven days.
- Use TED stockings or Flowtrons in first 24 hours post CS.

6. Postpartum care

- Frequency of clinical observations may be increased and if required modified on the MAADS chart.
- Encourage early mobilisation.
- Avoid dehydration.
- Perform skin integrity assessment and pressure area care as per the care plan.
- All woman with BMI >40 should receive thromboprophylaxis regardless of mode of delivery.
- All women with BMI >30 with other risk factors should be considered for pharmacological thromboprophylaxis regardless of the mode of delivery.
- For women with a high BMI (e.g. >30), particular consideration should be given to factors which may impact on the adequacy of the injection, including the site of administration, access to underlying muscle and the length of needle used.
- Women may require additional assistance with breast feeding. It has been shown that women with increased BMI have reduced initiation and duration of breast feeding.

- Before discharge from hospital discuss contraception. Discuss efficacy of various contraception methods.

6.1 Pre-conception counselling:

- The risk for falling pregnant in the future.
- Commencement of Folic Acid 5mg/day to decrease the risk of Neural Tube Defects (NTD).
- Delay pregnancy for at least 18 months if considering Bariatric surgery.
- Consider referral to Clinical Psychologist as postpartum depression has been reported to correlate positively with high BMI.

7. Legislative context

The following documents are required to give effect to this policy [i.e. the documents included are mandatory]:

- Health Department of WA (HDWA) [WA Health Consent to Treatment Policy MP0657/16](#)
- HDWA [Clinical Handover Policy](#)
- East Metropolitan Health Service (EMHS) [Bariatric Management Policy EMHS: 3](#)
- EMHS [Consent Policy EMHS: 5](#)
- EMHS [Patient Identification and Procedure Matching Policy EMHS: 21](#)

8. Supporting information

The following documents support the implementation of this policy:

- [Clinical Handover Procedure AKG-PRO-0521-19](#)

9. Relevant standards

National Safety and Quality Health Service (NSQHS) Standards (second edition):

- Standard 2 – Action 2.4:
- Standard 2 – Action 2.6:
- Standard 5 – Action 5.5:
- Standard 5 – Action 5.11:
- Standard 5 – Action 5.13:
- Standard 6 – Action 6.4:
- Standard 6 – Action 6.7:
- Standard 6 – Action 6.9:

10. References

1. Management of obesity in pregnancy RANZCOG guideline C-Obs 49.2021
2. National Institute for Health and Clinical Excellence. Weight management before, during and after pregnancy 2010.
3. KEMH. Clinical Guidelines 'Increased Body Mass Index : management of a woman with'-
[https://healthpoint.hdwa.health.wa.gov.au/policies/Policies/NMAHS/WNHS/WNHS.ORG.BodyMassIndex\(BMI\)morethan40-ManagementofaWomanwith.pdf](https://healthpoint.hdwa.health.wa.gov.au/policies/Policies/NMAHS/WNHS/WNHS.ORG.BodyMassIndex(BMI)morethan40-ManagementofaWomanwith.pdf)
4. IOM (Institute of Medicine) (US). Weight gain during pregnancy; re examining the guidelines. Washington DC: The National Academy Press; 2009.

<https://www.rcog.org.uk/globalassets/documents/guidelines/cmacercogjointguidelinemanagementwomenobesitypregnancya.pdf>

5. Baker P, Balen A, Poston L, Sattar N. Obesity and reproductive health - study group statement: consensus views arising from the 53rd study group: obesity and reproductive health. 2007 [cited 2009 August 17].
6. <https://www.rcog.org.uk/globalassets/documents/guidelines/cmacercogjointguidelinemanagementwomenobesitypregnancya.pdf>
7. [World Health organisation: Chapter 8, Overweight and obesity, \(high body mass index\)](#)
8. W. Philip at al <http://www.who.int/publications/cra/chapters/volume1/0497-0596.pdf>
9. Queensland Clinical Guidelines, Obesity in Pregnancy 2021
https://www.health.qld.gov.au/_data/assets/pdf_file/0019/142309/g-obesity.pdf

11. Glossary

The following definitions are relevant to this policy.

Term	Definition		
BMI	BMI is defined as weight in kilograms divided by the square of the height in metres (kg/m ²). Pre-pregnancy weight should be used where possible to avoid over estimation of BMI.		
WHO ⁶ Classification of BMI	Classification	BMI (kg/m²)	Risk of co-morbidities
	Underweight	Less than 18.5	Low (but risk of other clinical problems increased)
	Normal range	18.5- 24.9	Low
	Overweight	25-29.9	Low
	Obese I	30-34.9	Moderate
	Obese II	35-39.9	High
	Obese III	Greater than or equal to 40	High

12. Document owner

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13. Review

This guideline will be reviewed and evaluated as required to ensure relevance and currency. At a minimum it will be reviewed within one (1) year after first issue and at least every three (3) years thereafter.

Version	Effective from	Effective to	Amendment(s)
V4	03/10/2024	03/10/2027	Updated references Addition of weight management table

14. Authorisation

This guideline has been authorised and issued by the Director of Clinical Services.

Authorised by	Dr Tania Strickland – A/Director Clinical Services
Authorisation date	03/10/2024
Published date	03/10/2024